

## Project Design Phase-II

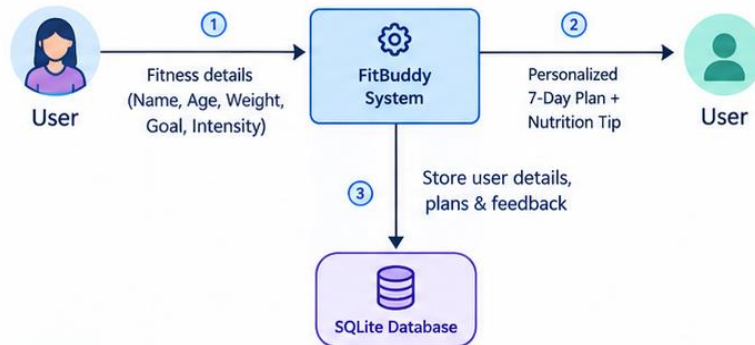
### Data Flow Diagram & User Stories

#### Project Name : FitBuddy

### Data Flow Diagrams – FitBuddy

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

#### Example: (Simplified)

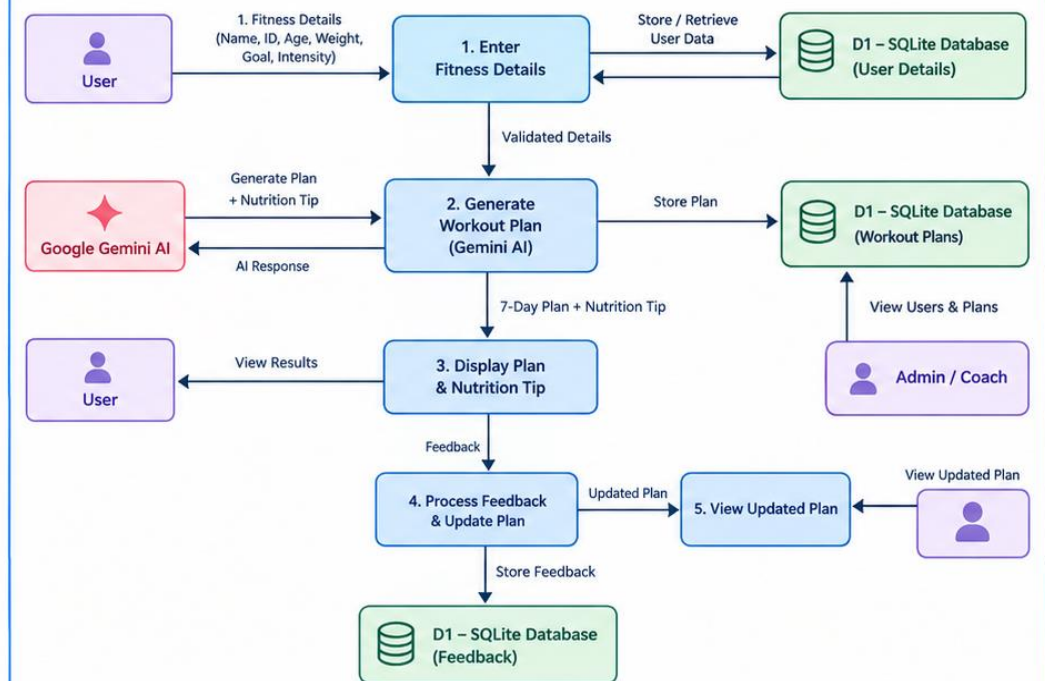


1. User enters fitness details (Name, Age, Weight, Goal, Intensity).
2. FitBuddy processes the data and generates a personalized 7-day workout plan with nutrition/recovery tip.
3. User details and plans are stored in the SQLite database.

#### User Stories

- As a user, I want to enter my fitness details so that FitBuddy can generate a personalized plan.
- As a user, I want to view my 7-day workout plan so that I can follow a structured routine.
- As a user, I want nutrition and recovery tips so that I can support my fitness goal.
- As a user, I want to submit feedback so that FitBuddy can update my plan.
- As an admin/coach, I want to view users and their plans so that I can monitor the information.

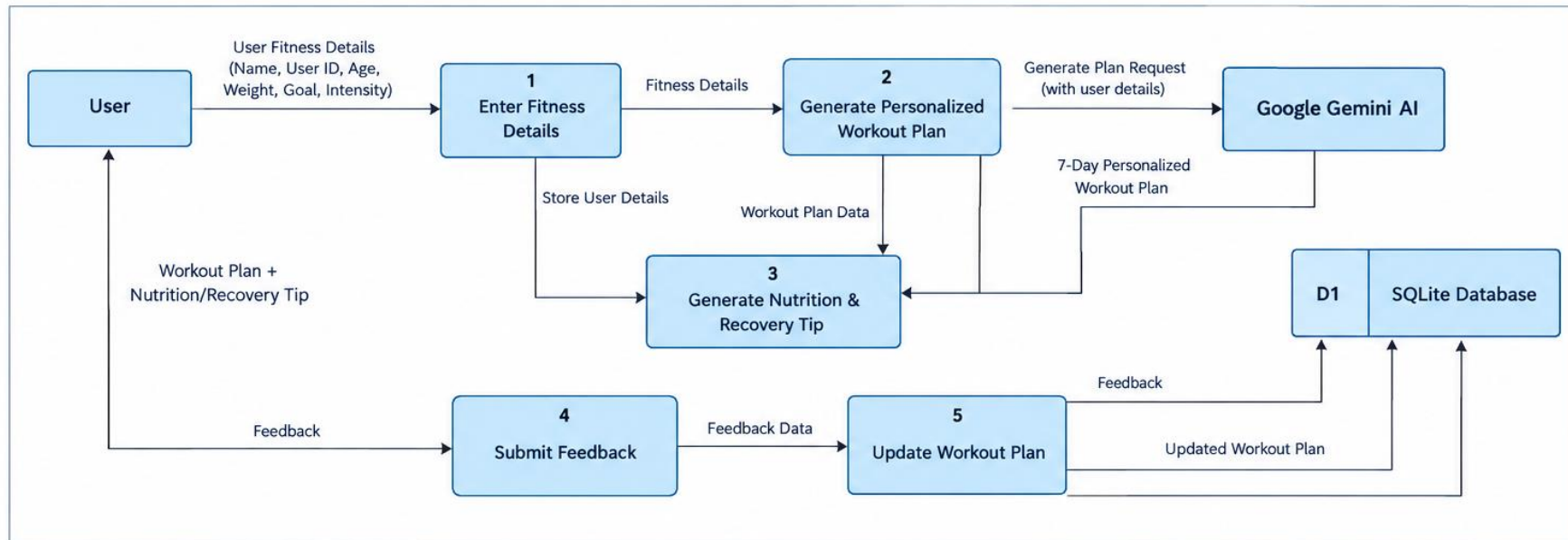
#### Example: DFD Level 0 (FitBuddy – System Overview)



## Data Flow Diagrams:

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### Example: FitBuddy – AI Fitness Plan Generator using Gemini Models



1. User enters their fitness details (Name, User ID, Age, Weight, Goal, Intensity) and the details are sent to the system.
2. The system processes the details and requests a personalized workout plan from Google Gemini AI.
3. Gemini AI returns the 7-day workout plan which is used to generate nutrition and recovery tips.
4. The generated workout plan and nutrition/recovery tip are displayed to the user.
5. The user can submit feedback, which is processed and used to update the workout plan. The user and plan information is stored in the SQLite database.